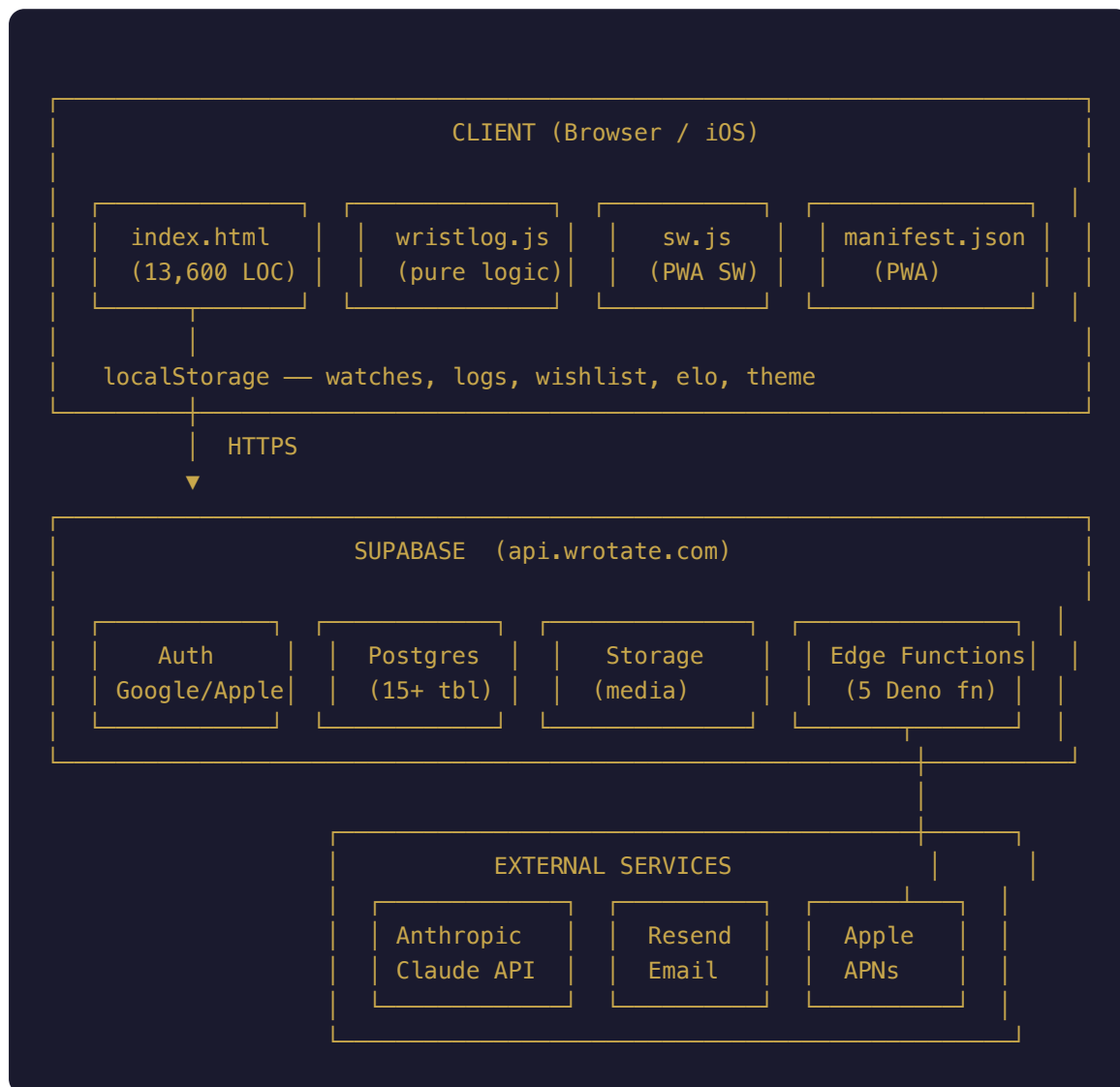


WRotate Architecture

Technical architecture document — March 2026

System Overview



Frontend

Stack: Vanilla JavaScript — no framework. Single HTML file (`index.html` , ~13,600 lines) containing all HTML, CSS, and JS. Pure business logic extracted to `wristlog.js` for testability.

External Dependencies (CDN)

Library	Version	Purpose
@supabase/supabase-js	v2	Auth, DB queries, storage, realtime
chart.js	v4.4.0	Statistics charts (deferred load)

Both loaded with SRI integrity hashes and `crossorigin="anonymous"` .

UI Views

View	Description
Feed	Social feed with wrist shots, likes, comments
Track	Daily wear logging, today's AI pick, history
Collection	Watch grid/list, ELO ranking game, tag filters
Stats	Charts, cost-per-wear, collection report
Wishlist	Drag-to-reorder wishlist with market prices
Profile	User profile, followers, showcase, settings
Clubs	Watch clubs, discovery, management

Theme System

CSS custom properties (`--bg` , `--surface` , `--gold` , etc.) with `[data-theme="dark"]` toggle. Theme applied synchronously before first paint via inline script.

Three modes: `Light` `Dark` `System`

Offline & Local State

Watches, logs, wishlist, and ELO ratings cached in `localStorage` . A pending-deletes queue (`wr_pendingDeletes`) survives offline and syncs on reconnect.

Backend — Supabase

Self-hosted Supabase instance at `api.wrotate.com` providing Auth, Postgres, Storage, Edge Functions, and Realtime.

Edge Functions (Deno)

Function	Trigger	Description
identify-watch	HTTP POST	Sends photo to Claude Vision API. Returns brand, model, reference, bounding box. Rate-limited: 100 req/user/hour.
search-watch-image	HTTP POST	Finds stock product photo from brand websites (40+ brand domain map). Falls back to user photo crop.
feedback-to-github	DB Webhook	Converts bug reports in feedback table to GitHub Issues for auto-fix pipeline.
send-email	DB Webhook	Sends email notifications via Resend API on new notifications rows. Respects per-user email prefs.
send-push	DB Webhook	Sends iOS push notifications via Apple APNs using ES256 JWT auth.

Database Schema

PostgreSQL via Supabase. All tables have Row Level Security (RLS) enabled.

Core Tables

Table	Purpose	Key Columns
profiles	User profiles	username, display_name, avatar_url, bio, theme_preference, email_prefs, collection_visibility, is_admin
watches	Watch collection	brand, name, reference_number, price, market_price, image_url, tags, elo_rating, straps
logs	Wear logs	watch_id, date, use_case, notes, photo_url, visibility
wishlist	Watch wishlist	brand, name, market_price, url, sort_order

Social Tables

Table	Purpose
follows	Follow relationships (follower_id, following_id)

follow_requests	Pending follow requests for private accounts
friend_requests	Close-friend mutual requests
likes	Log post likes
comments	Post comments with body text
comment_likes	Comment likes
notifications	All notification types (follow, comment, mention, club_invite, etc.)

Club & System Tables

Table	Purpose
clubs	Watch clubs (name, description, privacy)
club_members	Club membership (owner / member roles)
club_join_requests	Pending club join requests
club_invites	Direct club invitations
feedback	User bug reports and feature requests
device_tokens	iOS APNs push tokens
rate_limits	Per-user edge function rate limiting

Authentication

Supabase Auth with PKCE flow. Two OAuth providers:

- **Google OAuth** — Primary sign-in method
- **Apple Sign In** — With privaterelay email handling
- **Email/Password** — Dev/test accounts only (localhost-gated)

Auth state managed by Supabase JS client via `onAuthStateChange()` . On first OAuth login, a profile row is auto-created via `upsert` .

Storage

Single Supabase Storage bucket: `media`

- **User avatars** — uploaded via profile edit
- **Wear log photos** — uploaded via track/post flow
- **Watch images** — from stock photo search or user upload
- **Landing page assets** — screenshots under `landing/` prefix

Public CDN URLs via `api.wrotate.com/storage/v1/object/public/media/...`

AI Features

Watch Identification

`Claude Vision`

`Server-side`

User uploads a photo → `identify-watch` edge function sends base64 to Anthropic API → returns brand, model, reference, confidence, bounding box for each watch detected.

Stock Photo Search

`Web Scraping`

`Server-side`

After identification, `search-watch-image` tries product URLs from 40+ brand domains. Client falls back to canvas crop of user's original photo.

Daily Recommendation

`Algorithm`

`Client-side`

Scores watches by: idle time, day-of-week fit, weather, use-case history, ELO rating, and purchase recency. Suggests "today's pick."

ELO Ranking

`Algorithm`

`Client-side`

Head-to-head ranking game using standard ELO algorithm. Timed 3-second choices. Ratings persisted to Supabase.

PWA & iOS App

Service Worker

`sw.js` with stale-while-revalidate strategy for same-origin requests. Cross-origin requests (Supabase API, CDNs, OAuth) always bypass cache. Cache version manually bumped on deploys.

Native iOS App

Swift/Xcode project at `ios/Wrotate/` . WKWebView wrapper with native APNs push notification support. Device tokens registered in `device_tokens` table.

External Services

Service	Domain	Purpose
Supabase	<code>api.wrotate.com</code>	Auth, DB, Storage, Edge Functions
Anthropic	<code>api.anthropic.com</code>	Claude Vision AI (via edge function)
Resend	<code>api.resend.com</code>	Email notifications (via edge function)
Apple APNs	<code>api.push.apple.com</code>	iOS push notifications (via edge function)
GitHub	<code>api.github.com</code>	Auto-bug issue creation (via edge function)
wttr.in	<code>wttr.in</code>	Weather data for daily recommendation
jsDelivr CDN	<code>cdn.jsdelivr.net</code>	Supabase JS + Chart.js

CI/CD & Testing

GitHub Actions Workflows

Workflow	Trigger	Description
<code>test.yml</code>	Push / PR to main	Runs <code>vitest</code> test suite (575 tests across 17 files)
<code>auto-fix.yml</code>	Issue labeled <code>auto-bug</code>	Claude Code reads issue, edits code, runs tests, creates auto-merge PR

Test Coverage Areas

- ELO ranking algorithm
- Watch recommendation engine
- Statistics computation
- Data serialization (watch/log/wish)
- Save/edit watch logic
- Wear log operations
- Social features (follow, like, comment)
- Reviews (year/month)
- Utility functions (dates, money, HTML)
- Price formatting
- @mention parsing
- Username validation
- Network resilience / offline
- Auto-suggest tags

- Notification handling
- OEM strap guessing

Deployment

- **Frontend:** GitHub Pages at `wrotate.com` (custom domain via CNAME). Push to `main` auto-deploys.
- **Edge Functions:** Deployed via `supabase functions deploy <name> --project-ref xnzweevzrojmuozhpwzv --no-verify-jwt`
- **Database Migrations:** SQL scripts run manually in Supabase SQL Editor.
- **iOS App:** Built and submitted via Xcode. Bundle ID: `com.wrotate.Wrotate`